

MODULE 5

CTU AND CTD COUNTERS

PLC Siemens TIA Portal Course
Module 5: CTU and CTD Counters

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Level: Basic-Intermediate

Software: Siemens TIA Portal

Language: Ladder Diagram (LAD)

Introduction

In industrial automation it is very common to need to count events, parts produced, work cycles, or sensor activations. To do this, Siemens PLCs include counting instructions that allow accurate tracking of discrete processes.

Counters are fundamental in production lines, packaging systems, conveyor belts, assembly machines, and any process that requires counting units or events.

In this module you will learn to use the CTU and CTD counters in TIA Portal V20, understanding their internal operation, main variables, and real industrial applications.

What is a counter?

A counter is a programming instruction that increases or decreases a numeric value each time a given event is detected, usually a positive edge coming from a sensor or push button.

Industrial counters are widely used for:

- Counting parts produced.
- Controlling manufacturing batches.
- Managing machine cycles.
- Packaging systems.
- Automatic inventory control.
- Automating sequential processes.

CTU Counter (Count Up)

Definition

The CTU (Count Up) counter is an up counter. Every time it detects a positive edge on its counting input, it increases its current value by one unit.

When the accumulated value reaches the preset value, the counter output is activated.

Operation

- The counter starts at zero.
- A positive edge occurs on the counting input.
- The current value increases by one unit.
- The process continues until the programmed value is reached.
- When the reference value is reached, output Q is activated.

Industrial applications

- Counting boxes produced.
- Counting parts on a conveyor belt.
- Production batch control.
- Triggering alarms based on cycle count.
- Automatic packaging systems.
- Counting vehicles in parking lots.

Practical example

An alarm should be activated once 10 parts have passed in front of a photoelectric sensor.

Programmed value: 10 parts.

Result:

When the tenth part passes, the counter output is activated and triggers the alarm.

CTD Counter (Count Down)

Definition

The CTD (Count Down) counter is a down counter. It starts from a preset value and decreases by one unit each time it receives a pulse on the counting input.

When the current value reaches zero, the counter output is activated.

Operation

- The counter is loaded with an initial value.
- A positive edge is detected.
- The current value decreases by one unit.
- The process continues until it reaches zero.
- Output Q is activated.

Industrial applications

- Tracking remaining parts.
- Countdown of processes.
- Inventory management.
- Dosing systems.
- Maintenance cycle control.
- Packaging processes.

Practical example

A machine must stop when there are zero boxes left available.

Initial value: 20 boxes.

Result:

Each box removed decreases the counter until it reaches zero, at which point the machine stops automatically.

Important Siemens counter variables:

Variable	Description
CU (Count Up)	Up-counting input.
CD (Count Down)	Down-counting input.
R (Reset)	Resets the counter and returns the current value to zero.
LD (Load)	Loads an initial value into the down counter.
PV (Preset Value)	Reference or programmed value.
CV (Current Value)	Counter's current value.
Q	Counter output.

Comparison between CTU and CTD

Counter	Function
CTU	Up counting
CTD	Down counting

Positive Edge in Counters

Siemens counters work by detecting a positive edge.

This means counting only occurs when the signal changes from:

0 → 1

If the input remains in a high state, the counter will not continue increasing or decreasing.

For this reason, industrial applications are recommended to use:

- Photoelectric sensors.
- Inductive sensors.
- Limit switches.
- Push buttons with edge detection.

Real Industrial Applications

Production line

Count the quantity of parts manufactured.

Bottling plants

Stop the system after completing a production batch.

Storage systems

Control the number of boxes available.

Automatic parking lots

Count vehicles entering and leaving.

Maintenance systems

Generate alarms after a certain number of operating cycles.

Programming Best Practices

- Use descriptive names for each counter.
- Document the programmed value and its function.
- Use edge detection to avoid incorrect counts.
- Reset counters when appropriate.
- Verify the current value using watch tables.
- Test the program using PLCSIM before downloading it to the PLC.
- Avoid multiple simultaneous signals on the same counting input.

Practical Exercise

Design a program in TIA Portal V20 that allows you to:

Exercise 1

Count 20 boxes produced using a CTU counter.

Upon reaching 20 parts:

- Turn on a green lamp.
- Activate an alarm.

Exercise 2

Implement a CTD counter that starts at 15 parts.

Each time a part is removed:

- Decrease the counter.

When it reaches zero:

- Turn on a red light.
- Stop the machine.

Exercise 3

Create a daily production counting system that records:

- Quantity produced.
- Quantity remaining.
- End-of-batch alarm.

Conclusion

CTU and CTD counters are fundamental tools within Siemens PLC programming and allow production systems, inventory control, and automatic sequences to be implemented efficiently and reliably.

Mastering industrial counters is an essential step for any Siemens PLC programmer, since they are part of practically all modern industrial automation systems and allow professional solutions to be developed using TIA Portal V20.

END OF MODULE 5

AUTOMATION STUDIO X

PLC Siemens TIA Portal Training System